

Glossary

JEDEC: JEDEC (Joint Electron Device Engineering Council), Solid State Technology Association is the semiconductor engineering standardisation body of the Electronic Industries Alliance (EIA). JEDEC has been the foremost standards development organisation for the semiconductor industry since 1958.

Latency: Latency in memory is primarily a penalty cycle, since the information is spread through several rows of data fields and the strobe has to revisit the columns to move down to the next row. This column access usually involves two or three penalty cycles which are added to the normal access cycle. Therefore, you see a Column Access Strobe (CAS) delay of two (one normal + two penalty) or three (one normal + three penalty) cycles. Needless to say that CAS-2 is quite a bit faster than CAS-3.

meaty and usable bandwidth. Bandwidth essentially means the amount of data that can be transmitted in a given period of time (measured against a second). Whether a given bandwidth is sufficient or not depends on how much data needs to be transmitted. More bandwidth by itself is not necessarily a benefit. Everything needs to be synchronised to utilise the pathway, failing which it will simply clog some end.

Calculating the maximum theoretical bandwidth for memory technologies is fairly straightforward. SDRAM and DDR-SDRAM are systems that transfer data in parallel. The formula used to calculate the bandwidth for DDR-SDRAM is: Bandwidth = (bytes transferred per clock cycle) x (operations per clock cycle) x (clock cycles per second)

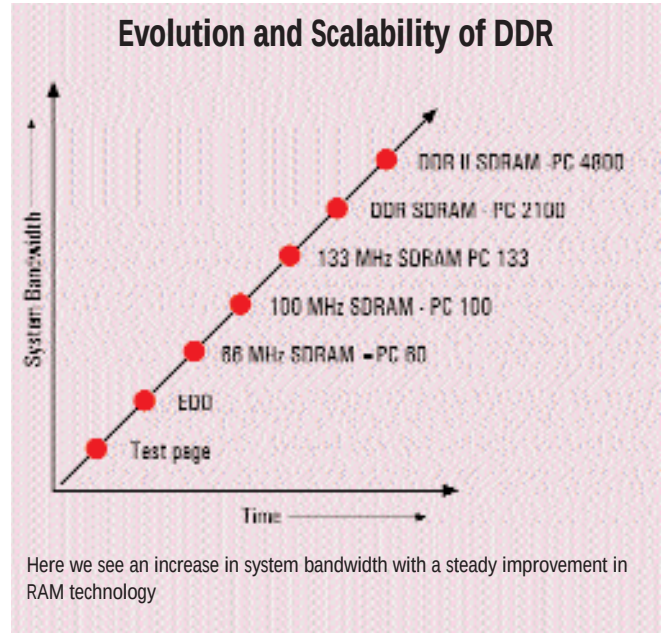
The bandwidth for Rambus RDRAM is calculated differently because the memory is serialised. Remember, RDRAM is 16-bit wide whereas SDRAM is 64-bit wide. For example, the ordinary PC100 SDRAM has the capability of transferring 800 MBps (8 bytes per cycle x 1 x 100 MHz). This was stretched to 1,064 MBps by PC133 simply by increasing the frequency.

As stated earlier, there are certain limitations to increasing frequency, the foremost limitation being cost. To combat this, DDR uses the unique 2n-prefetch logic to come up with a whopping 1.6 GBps for PC1600 (8 bytes per cycle x 2 x 100 MHz) and 2.1 GBps for PC2100. Compare this with RDRAM, and we find that under single-channel situations the fastest RDRAM (PC800) can come up with only 1.6 GBps (2 bytes per cycle x 2 x 400 MHz). Only under dual channel conditions does it double this bandwidth to a theoretical 3.2 GBps.

Though DDR too supports dual channel configurations, right now the only chipset solution it has is nVidia's nForce. The nVidia Crush chipset has support for dual channel DDR-SDRAM and, once available, it will offer the highest bandwidth offered by any solution, touching 4.2 GBps.

Waiting in queue

Waiting for something is bad enough, but waiting for something that you know is going to be late is worse. Latency is wasted time. It is the amount of time that a piece of hardware sits around doing nothing, waiting for another piece of hardware to deliver the goods for it to work on. In the PC scenario every single device is



apparently waiting for data. The bottlenecks are not the pathways that deliver data but the data mines themselves. Current memory, be it DDR, SDR or RDRAM, is a great deal slower than the processors of today and will be even slower than the processors of tomorrow. That is why CPU manufacturers have been forced to incorporate on-die cache and speculative prefetching. Otherwise, the processor would be sitting around most of the time waiting for the memory to get its act together.

So which memory solution is winning the waiting game? RDRAM seems to have problems when it comes to latency due to its serial nature. Since it only transmits 16 bits a cycle, it takes two cycles for each 32-bit WORD, as opposed to SDRAM's one cycle. Using two channels can effectively halve this, but you then need two RDRAM RIMMs and the waiting still continues. More importantly, since it functions in a serial fashion rather than as a parallel device like SDRAM, if there are multiple modules being used, all transactions must wind their way through all the modules between itself and the CPU before reaching its final destination. This slows down things further, more than in the case of the direct, parallel connection SDRAM. To top this, RDRAM deliberately slows down the responses of all the modules to the length of time it takes to communicate with the chip located further most. So the more modules you have, the worse the latency becomes.

So what's the upshot?

For several years, technological change in the world of memory has been evolutionary. The move from EDO to SDRAM, from one standard